Building Intelligent AI Agents — Participant Setup

Gheware UniGPS Solutions · 5-day hands-on workshop · local (self-hosted) Jupyter labs

Trainer: Rajesh Gheware **Tools:** Python · Jupyter · Hugging Face · LangChain / LangGraph · Ollama / Groq **Labs:** run locally — no browser/cloud lab provided

★ **Read first.** The workshop is 120 hands-on Jupyter labs run **locally**. The curriculum **requires** working with **MNIST**, a **real LLM** (agent labs) and **external APIs — Google Search & Wolfram Alpha** (Day 3): set these up. Every lab also has a **built-in offline fallback** — a **safety net** so a blocked key or port never strands anyone, **not** a reason to skip setup. Notebooks & answer keys are provided by the trainer.

0 · Hardware & OS

- **OS:** Windows 10/11, macOS or Linux (64-bit) · **RAM:** 8 GB min, **16 GB recommended** · **Disk:** ~10 GB free
- **Permissions:** install software; run local servers on `localhost` — ports `8888` (Jupyter), `11434` (Ollama, if used)

Tier 1 — Required core software

REQUIRED · EVERY LAB

- **Python 3.12** — **use this exact version** (mature wheels for every package incl. TensorFlow → no env troubleshooting) · a **virtual environment** (`venv`) · **Jupyter** (JupyterLab/Notebook or VS Code + Jupyter extension)

```
# create & activate a venv, then:
pip install --upgrade pip
pip install jupyterlab ipykernel nbformat numpy scikit-learn matplotlib tensorflow-cpu
```

`numpy/scikit-learn/matplotlib` → Days 1–2 · `tensorflow-cpu` → **Day 1 MNIST classifier** · Jupyter → run & grade every notebook.

Tier 2 — Required for the curriculum's named exercises

REQUIRED · PART OF THE COURSE

a. MNIST dataset (Day 1)

Framework is in Tier 1; participants must be able to **download MNIST** → allow `storage.googleapis.com`.

c. External APIs — Google Search & Wolfram (Day 3)

The "connect agents to external APIs" lab uses both — create free keys beforehand:

- **Google Serper** → `SERPER_API_KEY`
- **Wolfram Alpha** App ID → `WOLFRAM_ALPHA_APPID`

```
pip install wolframalpha  # Serper+Wolfram wrappers come from langchain-community
```

b. A real LLM for the agent labs (Days 3–5)

Pick **either** (both free, no paid account):

- **Option A — Groq API:** free account → key → `GROQ_API_KEY`
- **Option B — Ollama (local, no key):** install Ollama → `ollama pull llama3.2:1b` (~1.3 GB) → serves `localhost:11434`

```
pip install langchain langchain-community langchain-ollama langchain-groq langgraph
```

🔑 **BYOK — OpenAI (bring your own key).** The LLM labs run on **Groq / Ollama by default**. OpenAI is supported bring-your-own-key: set `OPENAI_API_KEY`, then `pip install openai` for the **Day-2 GPT text-gen** snippet (a commented reference in that lab), or `pip install langchain-openai` to use OpenAI as the **agent LLM (Days 3–5)**. *(OpenAI is paid/metered; Groq and Ollama are free — hence the default.)*

Tier 3 — Genuinely optional

NICE-TO-HAVE

Real Hugging Face transformers for the Day-2 "real BERT / GPT" demo cells — `pip install transformers torch`. *The Day-2 fine-tune lab has a complete, faithful offline path in scikit-learn; this only adds the real* `bert-tiny` *demo.*

🛡 **Built-in resilience (safety net — not a reason to skip setup).** Every lab degrades gracefully: **MNIST** blocked → offline 8×8 digits; **Serper/Wolfram** unavailable → deterministic local stand-in; **LLM** absent → deterministic mock. Set up the real services for the full intended experience.

Network / firewall allowlist (for participants' IT teams)

Purpose	Allow
pip / PyPI (all installs)	`pypi.org`, `files.pythonhosted.org`
MNIST dataset (Day 1)	`storage.googleapis.com`
Groq (LLM, Option A)	`console.groq.com`, `api.groq.com`
Ollama (LLM, Option B)	`ollama.com`, `registry.ollama.ai` (runtime local — `127.0.0.1:11434`)
Google Serper (Day 3)	`serper.dev`, `google.serper.dev`
Wolfram Alpha (Day 3)	`developer.wolframalpha.com`, `api.wolframalpha.com`
Hugging Face (optional)	`huggingface.co`, `cdn-lfs.huggingface.co`
OpenAI (optional, BYOK)	`platform.openai.com`, `api.openai.com`

Accounts / keys before the workshop

☐ **Groq** free account + `GROQ_API_KEY` *(or Ollama locally — no account)*
☐ **Serper.dev** free `SERPER_API_KEY` (Day 3)
☐ **Wolfram Alpha** `WOLFRAM_ALPHA_APPID` (Day 3)
☐ *(BYOK, optional)* **OpenAI** account + `OPENAI_API_KEY`

Set keys as environment variables (or a `.env` file the trainer points to).

What you do NOT need

- No GPU · no cloud/VM · no admin server
- No paid subscription **by default** (Groq, Ollama, Serper, Wolfram have free tiers)
- OpenAI is **optional (BYOK)**, not required
- No internet needed to **pass** a graded lab (offline fallbacks) — but Tier-2 services are needed to run the labs **as designed**

Pre-workshop 5-minute smoke test

```
python --version
python -c "import numpy, sklearn, matplotlib; print('sci stack OK')"
python -c "import tensorflow as tf; print('tensorflow', tf.__version__)"
python -c "import langchain, langchain_community, langgraph; print('langchain stack OK')"
jupyter lab --version
# LLM — run ONE of:
ollama run llama3.2:1b "say hi"                        # Option B (Ollama)
python -c "import langchain_groq; print('groq lib OK')"    # Option A (Groq)
# Keys present (Day 3 + LLM):
python -c "import os;[print(k,bool(os.getenv(k))) for k in ('GROQ_API_KEY','SERPER_API_KEY','WOLFRAM_ALPHA_APPID')]"
```